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Experimental investigation and thermodynamic assessment of La–Y–Ni ternary system in Ni-rich corner

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Abstract La–Y–Ni alloys exhibit high discharge capacity due to the formation of $AB_{3\text{-}3.8}$-type (A = La, Y; B = Ni) intermetallic compounds. However, the stable composition and temperature range for this type of phase are rarely reported, which restrains the development of La–Y–Ni hydrogen storage alloys with stable structure and high capacity. This paper focuses on the phase equilibria of the La–Y–Ni ternary system in Ni-rich corner. The phase constitution, microstructure, and equilibrated composition were experimentally determined at 1273 and 1148 K using X-ray diffraction (XRD), scanning electron microscopy (SEM), and energy-dispersive spectroscopy (EDS). The solubilities of La and Y in the binary compounds were measured. Two ternary compounds, 3R-LaY_2Ni_9 with the structure of $PuNi_3$ type and $La_{0.5}Y_{0.5}Ni_5$ with the structure of $CaCu_5$ type, existed at both temperatures. Based on the experimental data, the thermodynamic description of La–Y–Ni system was assessed by Calculation of Phase Diagram method. The calculated isothermal sections agree with the experimental data. The thermodynamic database is helpful for the design of La–Y–Ni hydrogen storage alloys.

Keywords La–Y–Ni system; Phase equilibria; Calculation of Phase Diagram (CALPHAD); Isothermal section; Phase transformation

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1 Introduction

AB_5-type and La–Mg–Ni-based hydrogen storage alloys are mainly used as anode materials of nickel–metal hydride (Ni–MH) batteries [1–5] due to their good electrochemical properties, high capacity and environment-friendliness. To date, the electrochemical discharge capacity of AB_5-type alloy reaches the limit of 350 $mAh\cdot g^{-1}$ and the preparation of La–Mg–Ni-based alloys is difficult because of the high volatility of Mg [6–11]. The superlattice La–Y–Ni-based hydrogen storage alloys without Mg are considered to be the potential alternative materials because of their higher theoretical discharge capacity ($\sim$ 400 $mAh\cdot g^{-1}$). The La–Y–Ni system contains several superlattice structures, such as AB_3-, A_2B_7- and A_5B_{19}-type phases, which are composed of [AB_5] and [A_2B_4] subunits stacking along with the *c* crystal axis in different atomic ratios of 1:1, 2:1 and 3:1 [12–15]. The [A_2B_4] subunit has two types of structures, C14 and C15, resulting in two structures of 2H and 3R respectively in the AB_3-, A_2B_7-, and A_5B_{19}-type phases [16]. However, the AB_3-, A_2B_7-, and A_5B_{19}-type phases are prone to hydrogen-induced amorphous (HIA) during hydrogen absorption and desorption processes [17, 18], which leads to the decay of discharge capacity and cycle life of electrode. The graphical representation of A_2B_4 and AB_5 stacking slabs in 2H and 3R AB_3-, A_2B_7-, A_5B_{19}-, and AB_4-type structures is shown in Fig. 1 [19, 20].

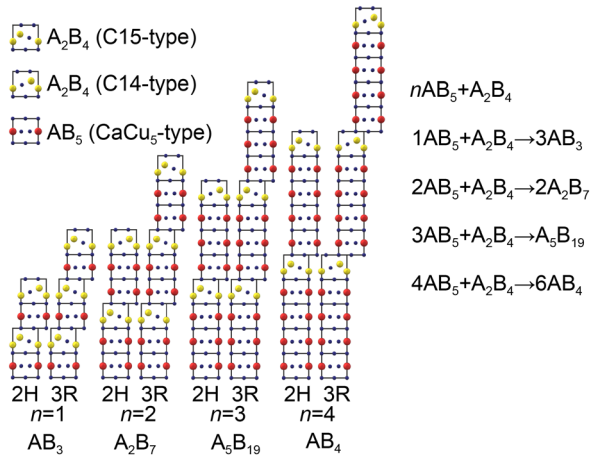

Fig. 1 Graphical representation of A_2B_4 and AB_5 stacking slabs in 2H and 3R AB_3-, A_2B_7-, A_5B_{19}- and AB_4-type structures

Previous researches have shown that the electrochemical performance of superlattice La–Y–Ni alloys can be improved by adjusting the element content [21–24]. Charbonnier et al. [25] found that 3R–Y_2Ni_7 phase had strong anti-amorphous ability and good cycle stability. Yan et al. [13] found that A_5B_{19}-type phase had better cyclic stability than the A_2B_7-type phase, which was attributed to the higher $[AB_5]/[A_2B_4]$ atomic ratio than that of A_2B_7-type. The AB_3-type phase has the lowest $[AB_5]/[A_2B_4]$ ratio (1:1), which results in the worst cycle stability. The A_5B_{19}-type phase with higher $[AB_5]/[A_2B_4]$ ratio than the AB_3- and A_2B_7-type phase can decrease the lattice strain of the alloys and decelerate the hydrogen-induced amorphous of the alloys in the hydrogenation/dehydrogenization process [13]. Therefore, the cyclic stability of alloys increases with the A_5B_{19} phases content. However, AB_3 phases exhibit higher discharge capacity compared with A_2B_7 and A_5B_{19} phases due to the higher content of A-side element. Thus, the electrode alloys with high discharge capacity can be obtained by adjusting the contents of those phases. The mutual substitution of La and Y affects the content and structural stability of AB_3-, A_2B_7- and A_5B_{19}-type phases. However, there are few literature reported the phase relations and thermodynamic stabilities of those key phases in the La–Y–Ni ternary system. Therefore, it is vital to explore the intermetallic compounds and their phase relations in the La–Y–Ni system concerning the Ni-rich corner.

For the three binary subsystems of the La–Y–Ni ternary system, La–Ni [26–29], Y–Ni [30–34], and La–Y [35, 36] have been previously investigated. The La–Ni system contains nine intermetallic compounds, $LaNi_5$, La_5Ni_{19}, La_2Ni_7, La_7Ni_3, $LaNi_3$, La_7Ni_{16}, La_2Ni_3, LaNi and La_3Ni. The existing temperature of the La_5Ni_{19} phase ranged from room temperature to 1276 K as reported by An et al. [29] and La–Ni binary phase diagram was thermodynamically assessed taking into account the La_5Ni_{19} phase. Subramanian and Smith [32] determined the enthalpy of formation of the nine Y–Ni intermediate compounds using electromotive force measurements. Considering the available experimental data, thermodynamic assessments were performed on the Y–Ni system by Du and Zhang [33]. The Y–Ni system contains nine intermetallic compounds, Y_3Ni, Y_3Ni_2, YNi, YNi_2, YNi_3, Y_2Ni_7, YNi_4, YNi_5 and Y_2Ni_{17}. The phase diagram of the La–Y binary system was first given by Spedding et al. [35] which included four solid solutions and one intermetallic compound, double hexagonal closest packing (dhcp)-La, face centered cubic (fcc)-La, body centered cubic (bcc)-LaY, hexagonal closest packed (hcp)-Y, and δ-LaY. The samarium-type δ-LaY has been found to be unstable by Wang et al. [36]. There is little information about the thermodynamic database of the La–Y system. With regard to La–Y–Ni ternary system, two ternary compounds $La_{0.5}Y_{0.5}Ni_5$ and LaY_2Ni_9 were found according to the available data [37, 38]. The accepted phase diagrams of La–Ni and Y–Ni systems are shown in Fig. 2, which are adopted in the present work.

This work focuses on the phase relations and thermodynamic stabilities of AB_3-, A_2B_7- and A_5B_{19}-type phases in the La–Y–Ni system. It is significant to establish phase equilibria at 1273 and 1148 K for alloy design in view of practical heat treatment temperature for the La–Y–Ni alloys [39–41]. The thermodynamic stabilities of superlattice compounds were firstly assessed using first-principles calculations. Then, the equilibrated phase constituents and compositions were experimentally investigated by X-ray diffraction (XRD), scanning electron microscopy (SEM), and energy-dispersive spectroscopy (EDS). Finally, the thermodynamic description of the La-Y-Ni system was optimized based on the experimental phase equilibria.

2 Experimental

High-purity metals of La (99.99 wt%), Y (99.99 wt%), and Ni (99.99 wt%) were used as the raw materials for samples preparation. The samples (about 50 g each) were prepared using an electric arc furnace with a non-consumable tungsten electrode and a water-cooled copper tray under a high pure argon atmosphere. A piece of Ti was placed in furnace and melted before the preparation of sample as an O_2 receiver. To ensure the composition homogeneity of samples, each sample was turned around and melted at least four times. The actual compositions of samples were measured by Optima 7300DV inductively coupled plasma atomic emission spectrometry (ICP).

The weight loss of the sample was less than 1 wt%. The number and compositions of all 19 samples are listed in Table 1. All the samples were wrapped in the tantalum foil

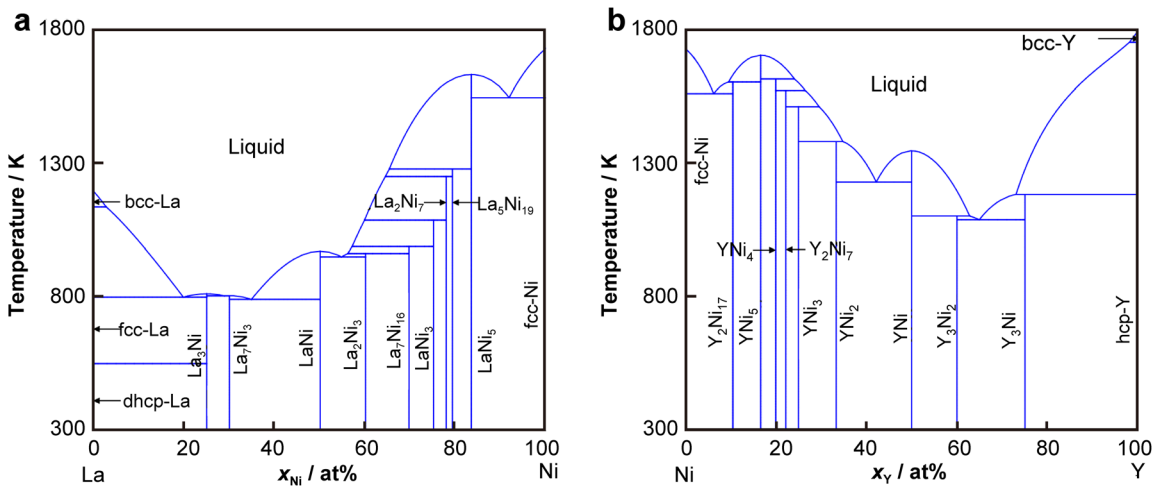

Fig. 2 **a** Calculated La-Ni phase diagram according to An et al. [29]; **b** Calculated Y–Ni phase diagram according to Du et al. [33]

and sealed in vacuumed quartz tubes. Then the samples were annealed at 1273 and 1148 K (with the temperature error of ± 5 K), respectively, for 24–48 h and cooled by water quenching.

The phase constitutions of annealed samples were identified by D2 PHASER XRD, with the scanning ranging from 10° to 80° at a rate of 2 (°)·min^{-1}. Rietveld refinement on the XRD patterns was carried out using TOPAS software. Microstructure and phase compositions of the samples were determined by JSM-7500 SEM and EDS. The Pandat software was used to perform the phase diagram calculations and optimizations [42].

3 Results and discussion

3.1 Phase equilibria of Ni-rich corner in La–Y–Ni ternary system at 1273 and 1148 K

Figure 3a–d shows the backscattered electron (BSE) images and the XRD patterns of samples #1–#3 annealed at 1273 K, respectively. The phases and the phase compositions are listed in Table 1. Grey $La_{0.5}Y_{0.5}Ni_5$ and dark grey fcc-Ni coexist in sample #1 (Fig. 3a). XRD pattern of sample #1 indicates that it consisted of fcc-Ni, $La_{0.5}Y_{0.5}Ni_5$ and Y_2Ni_{17} phases. Y_2Ni_{17} is not observed in the BSE image which may be due to its too low content. Therefore, it is considered that three-phase equilibrium fcc-Ni + $La_{0.5}Y_{0.5}Ni_5$ + Y_2Ni_{17} existed in sample #1. Sample #2 shows two-phase equilibrium of $La_{0.5}Y_{0.5}Ni_5$ + 3R-LaY_2Ni_9, as shown in Fig. 3b. Sample #3 consisted of $La_{0.5}Y_{0.5}Ni_5$ + YNi_5, where the YNi_5 and $La_{0.5}Y_{0.5}Ni_5$ show two different contrasts indicating that there is a solid solubility gap between them. The solubility of La in YNi_5 and $La_{0.5}Y_{0.5}Ni_5$ are 4.3 at% and 8.8 at%, respectively. It is hard to discriminate between the phase $La_{0.5}Y_{0.5}Ni_5$ and YNi_5 by XRD (Fig. 3d) because they have the same crystal structure. The difference between the two phases is discussed in the following section. The experimental phase equilibria of samples #1–#3 at 1273 K are shown in Fig. 3e.

Figure 4a–d shows the BSE images and XRD patterns of samples #4, #5 and #8 annealed at 1273 K, respectively. A single-phase of $LaNi_5$ presented in sample #4 with little impurity phase, as shown in Fig. 4a. The two-phase equilibrium of $LaNi_5$ + 2H-La_5Ni_{19} was observed in sample #5 from the BSE image (Fig. 4b). According to the XRD patterns shown in Fig. 4d, sample #5 is confirmed to be located in the 3R-LaY_2Ni_9 + 2H-La_5Ni_{19} + $LaNi_5$ three-phase region. 3R-LaY_2Ni_9 is not observed in the BSE image which may be due to the local composition deviating from the nominal composition. The solid solubility of Y in $LaNi_5$ is 8.2 at%.

The dark grey 2H-La_5Ni_{19} was observed in sample #8 from the BSE image (Fig. 4c) and XRD pattern (Fig. 4d). There is a light grey region (yellow arrow in Fig. 4c) in sample #8 which composition is about $La_{38.5}Y_{0.6}Ni_{60.9}$. The composition is close to the binary compound La_2Ni_3. However, the binary compound La_2Ni_3 would form through Liquid + La_7Ni_{16} → La_2Ni_3 at 961 K [29] in La–Ni system. Therefore, it is inferred that the light grey phase solidifies from the small amount of liquid during the quenching process. In addition, the thin layer compound which composition is close to La_2Ni_7 at the surface of 2H-La_5Ni_{19} maybe form through Liquid + La_5Ni_{19} → La_2Ni_7 at 1247 K [29]. The appearance of thin layered La_2Ni_7 in sample #8 supports the reaction that liquid reacted with 2H-La_5Ni_{19} during the cooling process. The compositions of annealed samples #6 and #7 are close to that of #8, which results in the same phenomenon. Thus, samples #6,

Table 1 Compositions and phase constitutions of La–Y–Ni samples annealed at 1273 and 1148 K

Sample No.	Actual composition / at%			Phases and compositions at 1273 K / at%				Phases and compositions at 1148 K / at%			
	La	Y	Ni	Phases	La	Y	Ni	Phase	La	Y	Ni
#1	5.81	5.16	89.03	fcc-Ni	0.4	0.1	99.5	fcc-Ni	0	0	100.0
				$La_{0.5}Y_{0.5}Ni_5$	7.5	8.9	83.6	$La_{0.5}Y_{0.5}Ni_5$	7.1	8.2	84.7
				Y_2Ni_{17}	–	–	–	Y_2Ni_{17}	–	–	–
#2	8.90	12.69	78.41	$La_{0.5}Y_{0.5}Ni_5$	8.8	10.6	80.6	$La_{0.5}Y_{0.5}Ni_5$	6.1	14.3	78.4
				3R-LaY_2Ni_9	9.6	14.2	76.2	3R-LaY_2Ni_9	9.5	14.9	75.6
#3	4.35	11.58	84.06	YNi_5	4.3	16.6	79.1	–	–	–	–
				$La_{0.5}Y_{0.5}Ni_5$	6.4	13.9	79.7	–	–	–	–
#4	12.95	7.09	79.96	$LaNi_5$	11.9	8.2	79.9	–	–	–	–
#5	15.76	6.04	78.20	$LaNi_5$	13.4	5.1	81.5	$LaNi_5$	–	–	–
				3R-LaY_2Ni_9	–	–	–	3R-LaY_2Ni_9	–	–	–
				2H-La_5Ni_{19}	14.5	8.6	76.9	2H-La_5Ni_{19}	18.2	5.5	76.3
#6	19.30	2.70	78.00	2H-La_5Ni_{19}	18.3	2.5	79.2	2H-La_2Ni_7	21.9	3.5	74.6
				2H-La_2Ni_7	27.1	1.7	71.2	YNi_2	33.2	0.9	65.9
				/	39.4	0.2	60.4	–	52.9	0	47.1
#7	19.90	3.80	76.30	2H-La_5Ni_{19}	16.9	4.3	78.8	–	–	–	–
				2H-La_2Ni_7	20.4	3.5	76.1	–	–	–	–
				–	38.9	0.3	60.8	–	–	–	–
#8	19.90	4.70	75.40	2H-La_5Ni_{19}	16.1	5.2	78.7	2H-La_2Ni_7	18.1	6.3	75.6
				2H-La_2Ni_7	19.3	4.3	76.4	YNi_2	27.9	4.6	67.5
				–	38.5	0.6	60.9	–	42.1	0	57.9
#9	3.02	17.77	79.20	YNi_5	1.6	18.5	79.9	YNi_5	1.4	17.6	81.0
				3R-Y_2Ni_7	2.1	21.9	76.0	3R-Y_2Ni_7	3.9	20.5	75.6
#10	4.10	20.70	75.20	3R-YNi_3	4.7	25.1	70.2	–	–	–	–
				3R-Y_2Ni_7	–	–	–	–	–	–	–
#11	3.27	22.15	74.58	3R-YNi_3	2.8	25.5	71.7	3R-YNi_3	3.0	27.4	69.6
				3R-Y_2Ni_7	2.7	22.9	74.4	3R-Y_2Ni_7	–	–	–
#12	4.80	20.70	74.50	3R-LaY_2Ni_9	4.8	20.1	75.1	–	–	–	–
				3R-Y_2Ni_7	–	–	–	–	–	–	–
#13	3.13	26.46	70.40	YNi_2	4.3	31.1	64.6	YNi_2	3.6	33.1	63.3
				3R-YNi_3	2.2	26.3	71.5	3R-YNi_3	0.6	28.2	71.2
#14	6.10	22.30	71.60	YNi_2	13.5	18.0	68.5	–	–	–	–
				3R-YNi_3	3.8	21.7	74.5	–	–	–	–
#15	14.03	17.81	68.15	3R-LaY_2Ni_9	7.3	20.2	72.5	3R-LaY_2Ni_9	8.7	20.5	70.8
				YNi_2	19.7	13.9	66.4	YNi_2	19.4	16.4	64.2
#16	20.36	13.22	66.42	3R-LaY_2Ni_9	11.6	16.1	72.3	–	–	–	–
				YNi_2	24.1	11.1	64.8	–	–	–	–
				Liquid	51.3	0	48.7	–	–	–	–
#17	13.24	27.81	58.95	[a]	–	–	–	–	–	–	–
#18	32.82	10.64	56.55	[a]	–	–	–	–	–	–	–

[a]Sample melted during heat treatment

#7 and #8 are possibly located in the two-phase equilibrium region of Liquid + 2H-La_5Ni_{19} at 1273 K. The maximum solubility of Y in 2H-La_5Ni_{19} would obtain from sample #5, which is about 8.6 at%. The experimental phase equilibria of samples #4, #5 and #8 at 1273 K are shown in Fig. 4e.

Figure 5a–e shows the BSE images and XRD patterns of samples #9–#12 annealed at 1273 K, respectively. BSE

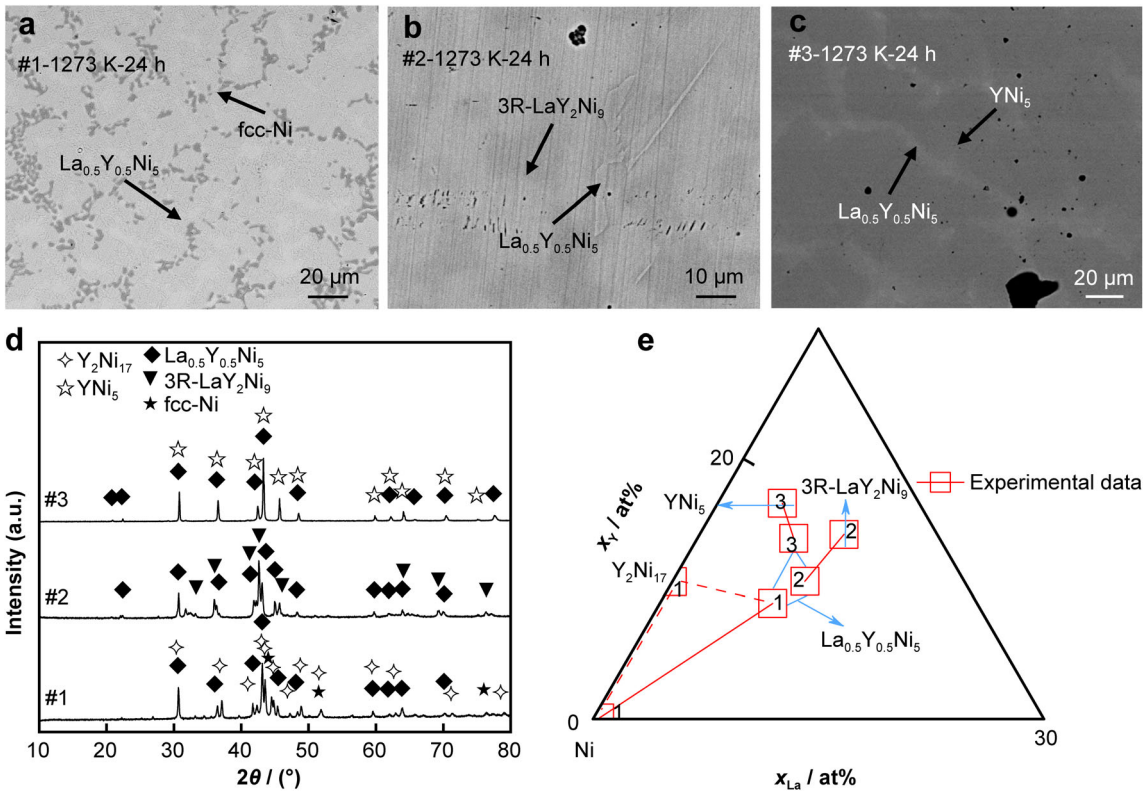

Fig. 3 **a**–**c** BSE images, **d** XRD patterns and **e** experimental phase equilibria of samples #1–#3 annealed at 1273 K

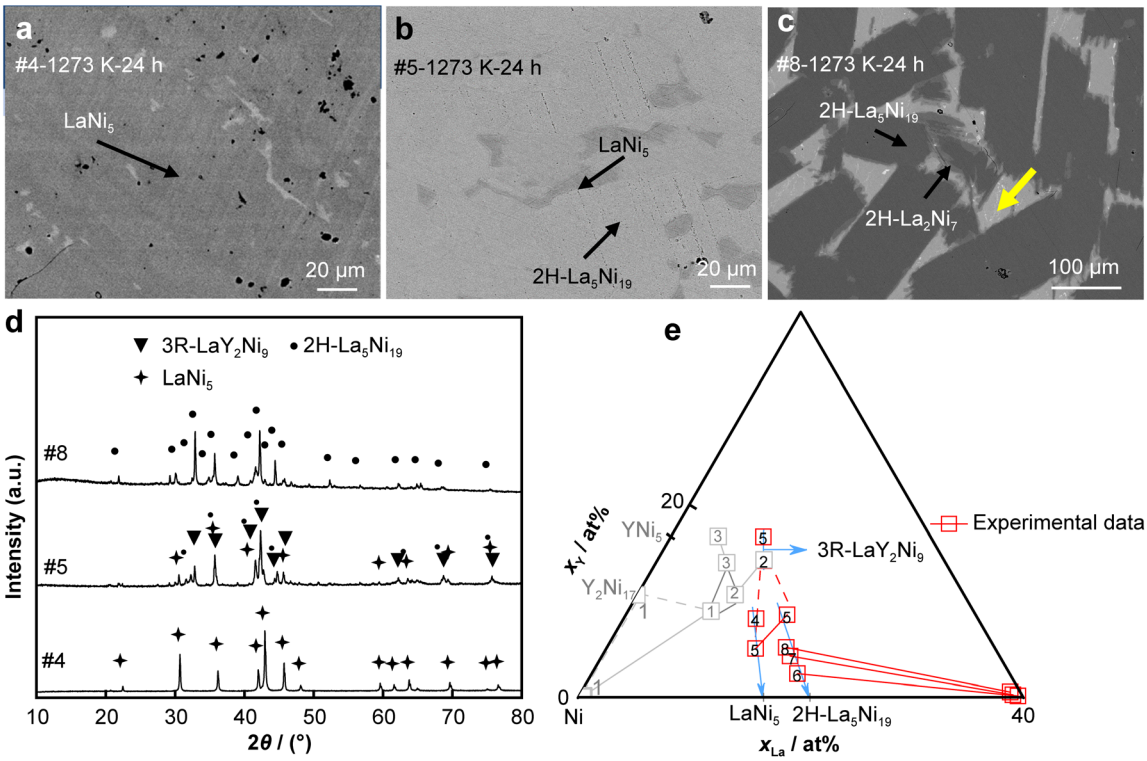

Fig. 4 **a**–**c** BSE images, **d** XRD patterns and **e** experimental phase equilibria of samples #4, #5 and #8 annealed at 1273 K

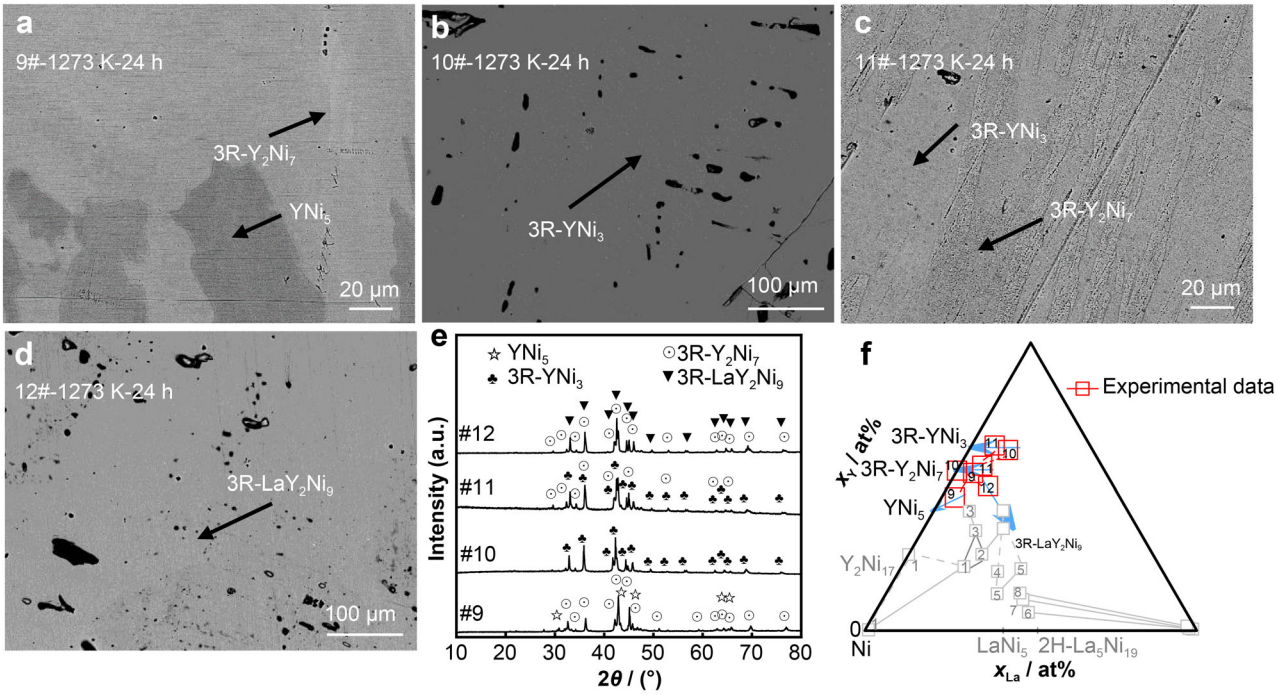

Fig. 5 **a**–**d** BSE images, **e** XRD patterns and **f** experimental phase equilibria of samples #9–#12 annealed at 1273 K

image (Fig. 5a) and XRD pattern (Fig. 5e) of sample #9 indicate that the sample is in the two-phase region of YNi_5 + 3R-Y_2Ni_7. Nearly single-phase 3R-YNi_3 was observed in sample #10 (Fig. 5b). Sample #11 is composed of 3R-YNi_3 and 3R-Y_2Ni_7. However, because of the little composition difference between the compounds 3R-YNi_3 and 3R-Y_2Ni_7, it is hard to observe the contrast difference from Fig. 5c. XRD pattern of sample #12 displays the characteristic peaks of 3R-LaY_2Ni_9 and 3R-Y_2Ni_7, but 3R-Y_2Ni_7 was not observed in the BSE image (Fig. 5d). The maximum solid solubilities of La in 3R-YNi_3 and 3R-Y_2Ni_7 are 4.7 at% and 2.7 at%, respectively. The experimental phase equilibria of samples #9–#12 at 1273 K are shown in Fig. 5f.

Figure 6a–e shows the BSE images and XRD patterns of samples #13–#16 annealed at 1273 K, respectively. Samples #13 and #14 annealed at 1273 K are both composed of YNi_2 + 3R-YNi_3. Grey YNi_2 and dark grey 3R-LaY_2Ni_9 coexist in sample #15 (Fig. 6c). The appearance of Y phase in sample #15 may be caused by uneven smelted Y particles. BSE image (Fig. 6d) and XRD pattern (Fig. 6e) of sample #16 indicate that the sample is composed of YNi_2 + 3R-LaY_2Ni_9. In addition, there is a light grey region (yellow arrow in Fig. 6d) in sample #16 with composition of $La_{51.3}Ni_{48.7}$ determined by EDS. It is close to the binary compound LaNi. However, the binary compound LaNi forms through Liquid + La_2Ni_3 → LaNi at 948 K [29] in La–Ni system. Therefore, it is regarded that the light grey region in sample #16 is formed from the liquid during the cooling process. Hence, sample #16 is probably located in the Liquid + YNi_2 + 3R-LaY_2Ni_9 equilibrium region at 1273 K. The maximum solubility of La in YNi_2 is 24.1 at%. Both samples #17 and #18 completely melted during annealing at 1273 K, which was not examined in this work. The experimental phase equilibria of samples #13–#16 at 1273 K are shown in Fig. 6f.

To further characterize the La-Y-Ni ternary system, the phase equilibria at 1148 K were also experimentally determined using equilibrated alloys. Table 1 shows the EDS results of samples #1, #2, #5, #6, #8, #9, #11, #13 and #15 annealed at 1148 K. The determined phase equilibria of samples at 1148 K are the same with that at 1273 K except samples #6 and #8. Figure 7 shows the BSE image of sample #8 and XRD patterns of samples #6 and #8 annealed at 1148 K. The dark grey 2H-La_2Ni_7 was observed in sample #8 from the BSE image. There is a light grey region (yellow arrow in Fig. 7a) in sample #8 which composition is about $La_{47.5}Ni_{52.5}$. Its composition is close to the binary compound LaNi. The light grey phase solidifies from the small amount of liquid during the quenching process. In addition, the thin layer compounds $LaNi_2$ at the surface of 2H-La_2Ni_7. The appearance of $LaNi_2$ in sample #8 supports the reaction that liquid reacted with 2H-La_2Ni_7 during the cooling process. The compositions of annealed sample #6 are close to that of #8, which results in the same phenomenon. Thus, samples #6 and #8 are possibly located in the two-phase equilibrium region of Liquid + 2H-La_2Ni_7 at 1148 K.

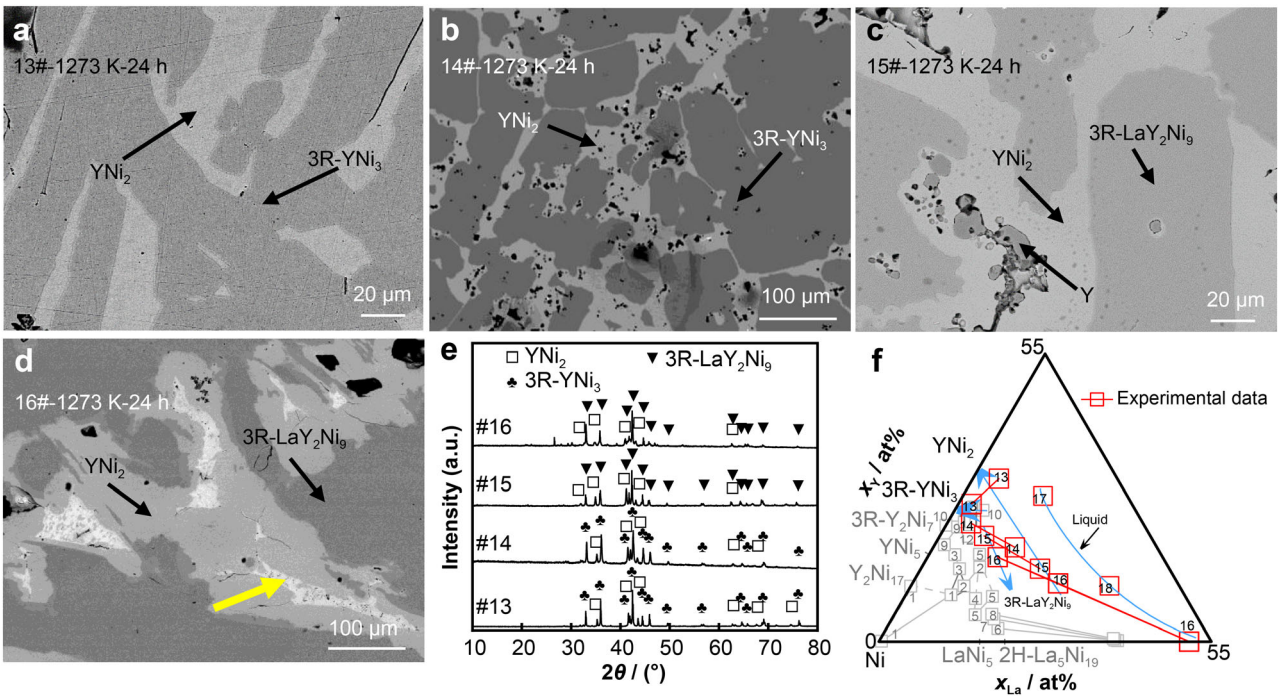

Fig. 6 **a**–**d** BSE images, **e** XRD patterns and **f** experimental phase equilibria of samples #13–#16 annealed at 1273 K

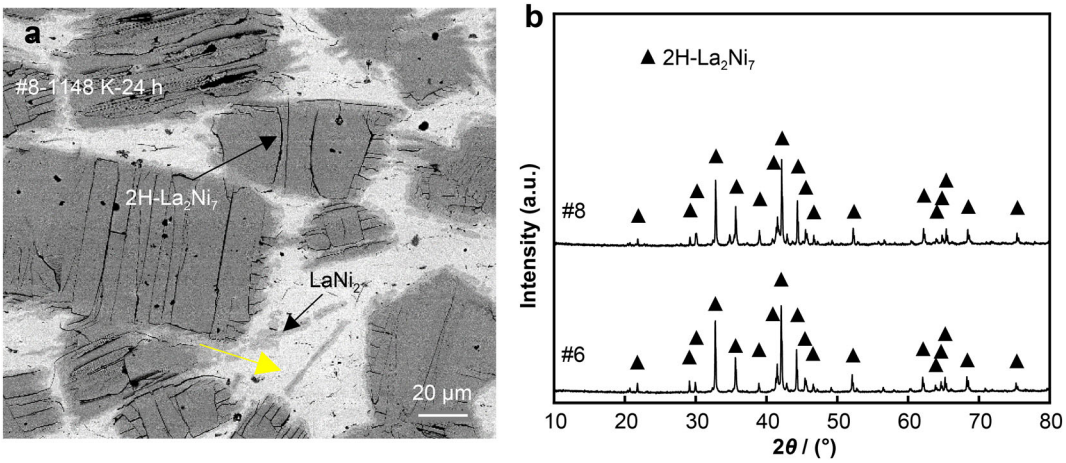

Fig. 7 **a** BSE image of sample #8 and **b** XRD patterns of samples #6 and #8 annealed at 1148 K

At 1148 K, the maximum solid solubilities of Y in 2H-La_5Ni_{19} and 2H-La_2Ni_7 are 5.5 at% and 6.3 at%, and the maximum solid solubilities of La in YNi_5, 3R-Y_2Ni_7, 3R-YNi_3 and YNi_2 are 1.4 at%, 3.9 at%, 3.0 at% and 19.4 at%, which are smaller than that at 1273 K. The solubility deviation is attributed to the heat treatment temperature. According to the present work, the determined phase equilibrium at 1148 K is different from that at 1273 K, where 2H-La_2Ni_7 phase is stable in La–Y–Ni system at 1148 K.

The partial experimental phase diagrams of the La–Y–Ni system at 1273 and 1148 K were constructed based on the experimental data obtained in this study, as shown in Fig. 8. The composition of binary $LaNi_5$, La_5Ni_{19}, La_2Ni_7, YNi_2, YNi_3, Y_2Ni_7, YNi_4, YNi_5 and Y_2Ni_{17} compounds come from those binary systems at 1273 and 1148 K, respectively.

3.2 Composition and crystal structure difference between $LaNi_3$, YNi_3 and LaY_2Ni_9

All of the intermetallic compounds $LaNi_3$, YNi_3 and LaY_2Ni_9 are rhombohedral $PuNi_3$-type structure (space group $R\bar{3}m$). The $PuNi_3$-type structure can be described as

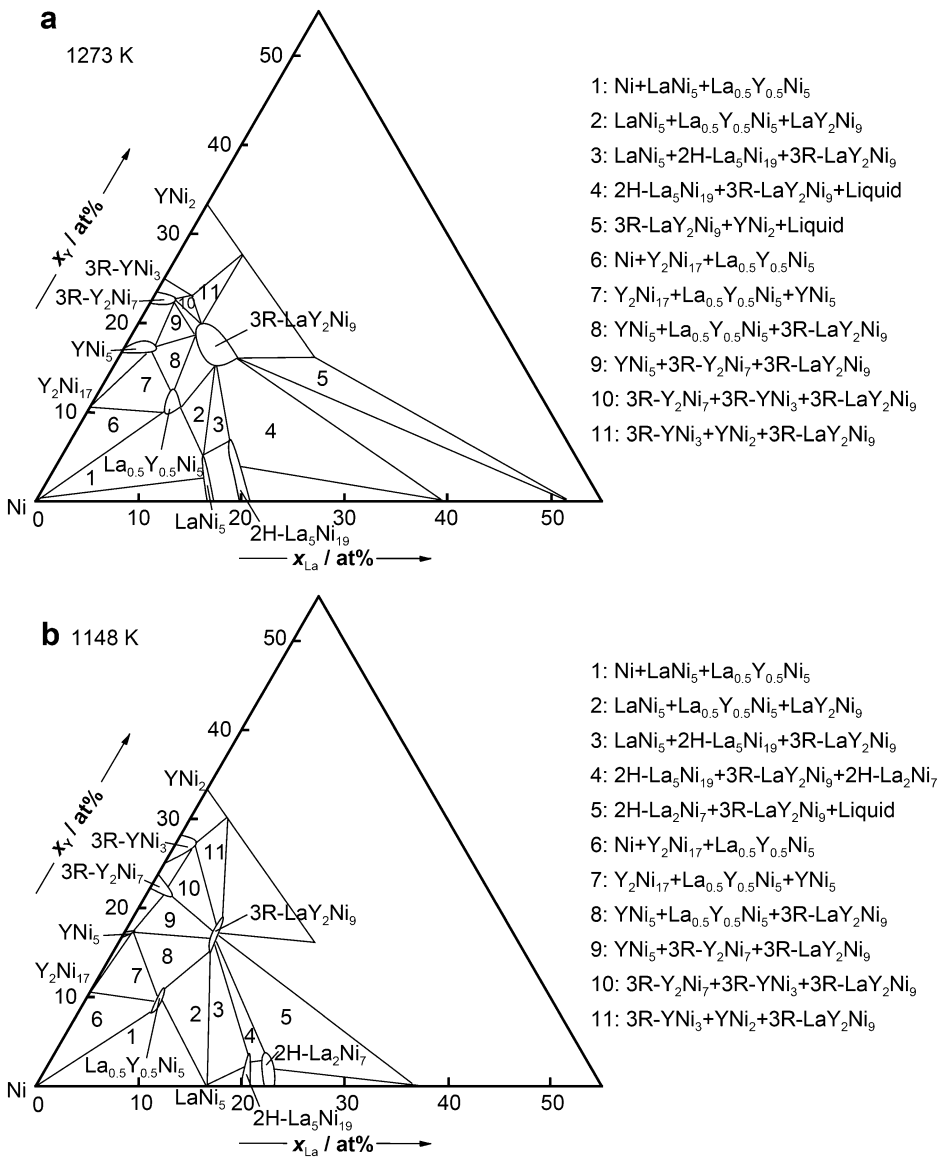

Fig. 8 Partial experimental phase diagram of La–Y–Ni system at **a** 1273 K and **b** 1148 K

an intergrowth of RM_5 (Haücke phase) and RM_2 (Laves phase) type structures [43]. The binary compound $LaNi_3$ forms through Liquid + La_2Ni_7 → $LaNi_3$ at 1084 K [29]. It was not found in the equilibrium at 1148 and 1273 K based on present experimental data. 3R-YNi_3 and 3R-LaY_2Ni_9 can exist in the equilibrium at 1148 and 1273 K. The atomic parameters of 3R-YNi_3 and 3R-LaY_2Ni_9 phases are listed in Table 2, which are obtained from studies by Latroche et al. [38] and Baddour-Hadjean et al. [43]. The diffraction peak locations of 3R-YNi phase and 3R-LaY_2Ni_9 phase are extremely close, it is hard to distinguish the two phases based on XRD patterns analysis. As samples #10, #11, #13 and #14 contain 3R-YNi_3 and samples #12 and #15 contain 3R-LaY_2Ni_9, their XRD patterns were analyzed carefully.

Figure 9 plots the Rietveld refinement of samples #11–#15 annealed at 1273 K, and the lattice parameters are listed in Table 3. The substitution of La for Y would lead to a corresponding increase in the lattice constant and the unit cell volume due to the atomic radius of La (0.187 nm) element is larger than that of Y (0.180 nm). The lattice constants *a* and *c* and the unit cell volume (*V*) of 3R-YNi_3 in samples #10, #11, #13 and #14 increase with the increase of La content. Compared with sample #15, the lattice constants and the unit cell volume of 3R-LaY_2Ni_9 in sample #12 have been decreased. The possible reason is that the substitution of Y for La leads to a corresponding decrease in the lattice parameters. The 3R-LaY_2Ni_9 phase with a larger cell unit volume which can provide more available interstices for H in the lattice exhibited a slightly

Table 2 Crystallographic data for 3R-YNi_3 and 3R-LaY_2Ni_9 [38, 43]

Phase	YNi_3			
Prototype/pearson symbol	$PuNi_3$, *hR*36			
Lattice parameter	*a* = 0.49779 nm, *c* = 2.44493 nm			
Atoms	Site	*x*	*y*	*z*
Y1	3a	0	0	0
Y2	6c	0	0	0.13864
Ni1	3b	0	0	0.50000
Ni2	6c	0	0	0.33364
Ni3	18h	0.5	0.5	0.08162
Phase	LaY_2Ni_9			
Prototype/pearson symbol	Mg_2LaNi_9, *hR*36			
Lattice parameter	*a* = 0.50343 nm, *c* = 2.45071 nm			
Atoms	Site	*x*	*y*	*z*
La	3a	0	0	0
Y	6c	0	0	0.13971
Ni1	3b	0	0	0.50000
Ni2	6c	0	0	0.33193
Ni3	18h	0.5	0.5	0.08242

higher maximum discharge capacity (258 mAh·g^{-1}) than the 3R-YNi_3 (238 mAh·g^{-1}) phase [44, 45].

3.3 Composition difference between YNi_5 and $La_{0.5}Y_{0.5}Ni_5$ phases

From the data on crystal structure reported by Mendelsohn et al. [37] and Sorgic et al. [46], it can be found that $La_{0.5}Y_{0.5}Ni_5$ showed the same structure as YNi_5, whose crystal structure parameters are listed in Table 4. The grey $La_{0.5}Y_{0.5}Ni_5$ and the dark grey YNi_5 coexist in sample #3 (Fig. 3c). The content difference between YNi_5 and $La_{0.5}Y_{0.5}Ni_5$ can be easily noticed. The XRD pattern of sample #3 with the enlarged view of two characteristic peaks is shown in Fig. 10. Although YNi_5 and $La_{0.5}Y_{0.5}Ni_5$ have the same structure, their XRD peaks are not completely same due to the different lattice parameters caused by different La solid solubility. In other words, there is a miscibility gap between YNi_5 and $La_{0.5}Y_{0.5}Ni_5$. Therefore, YNi_5 and $La_{0.5}Y_{0.5}Ni_5$ are two different phases and $La_{0.5}Y_{0.5}Ni_5$ phase is a stable ternary compound in La-Y-Ni system.

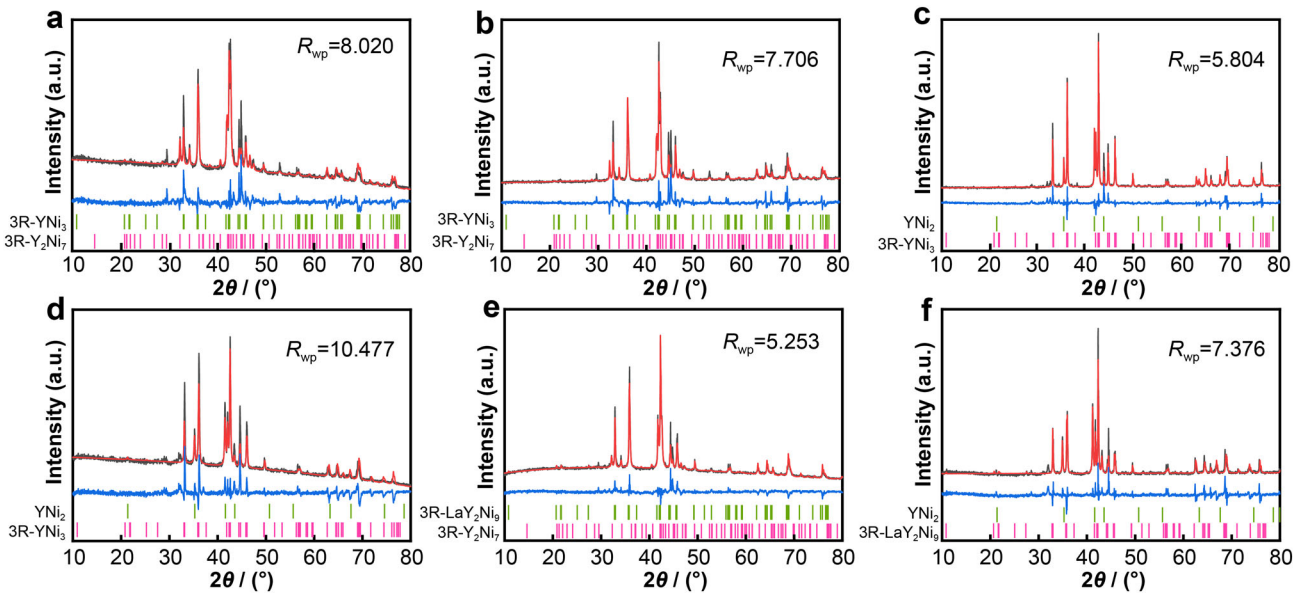

Fig. 9 Rietveld refinement patterns for samples annealed at 1273 K: **a** #10, **b** #11, **c** #13, **d** #14, **e** #12 and **f** #15 (R_{wp}: *R*-weighted pattern factor)

Table 3 Lattice parameters for 3R-YNi_3 and 3R-LaY_2Ni_9 from Rietveld refinement of XRD data

Sample No.	Phase	Composition / at%			Lattice parameters			Phase abundance / wt%
		La	Y	Ni	*a* / nm	*c* / nm	*V* / nm^3	
#10	3R-YNi_3	4.7	25.1	70.2	0.4987	2.4367	0.5247	43.76
#11	3R-YNi_3	2.8	25.5	71.7	0.4979	2.4363	0.5230	59.38
#13	3R-YNi_3	2.2	26.3	71.5	0.4980	2.4329	0.5225	81.77
#14	3R-YNi_3	3.8	21.7	74.5	0.4981	2.4332	0.5228	81.86
#12	3R-LaY_2Ni_9	4.8	20.1	75.1	0.5011	2.4473	0.5323	64.80
#15	3R-LaY_2Ni_9	7.3	20.2	72.5	0.5012	2.4418	0.5312	69.05

Table 4 Crystallographic data of YNi_5 and $La_{0.5}Y_{0.5}Ni_5$ [37, 46]

Phase	Prototype	Pearson symbol	Space group	Lattice parameters / nm		
				a	*b*	*c*
YNi_5	$CaCu_5$	*hP*6	*P*6/*mmm*	0.4883	0.4883	0.3967
$La_{0.5}Y_{0.5}Ni_5$	$CaCu_5$	*hP*6	*P*6/*mmm*	0.4932	0.4932	0.4006

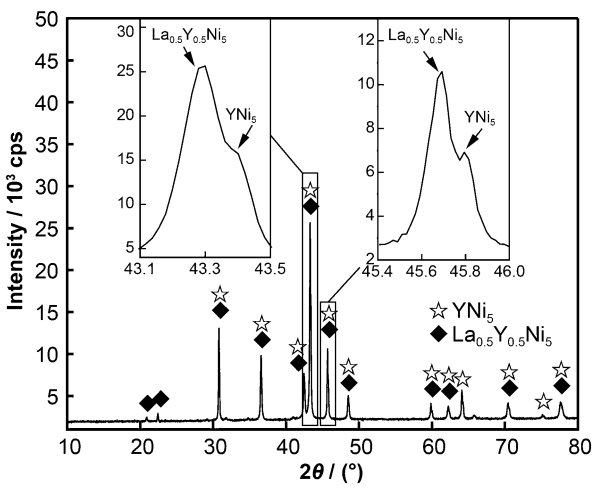

Fig. 10 XRD pattern of sample #3 annealed at 1273 K

3.4 Thermodynamic assessment of La–Y–Ni system

The Gibbs free energies for pure La, Y and Ni are taken from the Scientific Group Thermodata Europe (SGTE) database (Version 5.0) [47]. bcc, fcc, hcp, dhcp, and liquid phases are described using the substitutional solution model as follows:

$$G^{\varphi} = \sum_i x_i G_i^{\circ,\varphi} + RT \sum_i x_i \ln x_i + G^{\mathrm{ex},\varphi} \quad (1)$$

where x_i represents the mole fraction of element i, $G_i^{\circ,\varphi}$ is the molar Gibbs energy of element i as φ phase, R is the gas constant, T is the absolute temperature and $G^{\mathrm{ex},\varphi}$ is the excess Gibbs free energy of the phase, which is defined by an equation of the following form:

$$G^{\mathrm{ex},\varphi} = x_{\mathrm{La}}x_{\mathrm{Ni}} \sum_j {}^{j}L^{\varphi}_{\mathrm{La,Ni}}(x_{\mathrm{La}} - x_{\mathrm{Ni}})^j + x_{\mathrm{Y}}x_{\mathrm{Ni}} \sum_j {}^{j}L^{\varphi}_{\mathrm{Y,Ni}}(x_{\mathrm{Y}} - x_{\mathrm{Ni}})^j + x_{\mathrm{La}}x_{\mathrm{Y}} \sum_j {}^{j}L^{\varphi}_{\mathrm{La,Y}}(x_{\mathrm{La}} - x_{\mathrm{Y}})^j \quad (2)$$

where ${}^{j}L^{\varphi}_{\mathrm{La,Ni}}$, ${}^{j}L^{\varphi}_{\mathrm{Y,Ni}}$ and ${}^{j}L^{\varphi}_{\mathrm{La,Y}}$ are the binary interaction parameters.

The Gibbs energy of binary phases $LaNi_5$, 2H-La_5Ni_{19}, 2H-La_2Ni_7, YNi_5, 3R-Y_2Ni_7, 3R-YNi_3 and YNi_2 are presented by the compound energy formalism (CEF) which considered the solubility of La or Y. The solubility of Ni in $La_{0.5}Y_{0.5}Ni_5$ ranges from 79.7 at% to 83.6 at%, which is relatively low compared with the composition of Ni in the compound. The solubility of La in $La_{0.5}Y_{0.5}Ni_5$ ranges from 6.4 at% to 8.8 at%. The solubility of Y in $La_{0.5}Y_{0.5}Ni_5$ ranges from 8.9 at% to 13.9 at%, which shows a large deviation from its stoichiometric ratio. In order to simplify the sublattice model of phase, $La_{0.5}Y_{0.5}Ni_5$ phase in the La–Y–Ni system is modeled as $(La,Y)_1(La,Y)_1Ni_{10}$. The solid solution range of La, Y and Ni in 3R-LaY_2Ni_9 is larger than that of $La_{0.5}Y_{0.5}Ni_5$, thus 3R-LaY_2Ni_9 phase in the La–Y–Ni system is modeled as $(La,Y,Ni)_1(La,Y,Ni)_2(La,Y,Ni)_9$.

Other binary intermetallic phases, such as $LaNi_3$, La_7Ni_{16}, La_2Ni_3, LaNi, La_7Ni_3, La_3Ni, Y_2Ni_{17}, YNi_4, YNi, Y_3Ni_2, Y_3Ni, are deemed to be stoichiometric compounds because the solid solution of the third element in these

compounds can be ignored. The Gibbs energy function for stoichiometric compounds is described as:

$$G^{La_xY_yNi_z} = xG_{La}^{0,dhcp} + yG_{Y}^{0,hcp} + zG_{Ni}^{0,fcc} + a - bT \quad (3)$$

where $G_{La}^{0,dhcp}$, $G_{Y}^{0,hcp}$ and $G_{Ni}^{0,fcc}$ are Gibbs energies of components in their reference states, x, y and z are the mole fractions of La, Y and Ni, the parameters a and b are the formation enthalpy and entropy of the compounds.

Because of the limited data on enthalpies of formation (ΔH_f), the ΔH_f for end-members of those binary phases in La–Y–Ni system were obtained using first-principles calculations based on the density functional theory (DFT). The calculations were performed by the Cambridge Serial Total Energy Package (CASTEP) [48]. The generalized gradient approximation based on the Perdew-Burke-Ernzerhof (PBE) [49] method was employed to acquire accurate correlation energies. The plane-wave cut-off energy and the k-point separation were 400 eV and 0.3 nm^{-1}, respectively. Both the unit cell sizes and the ionic coordinates were fully relaxed to find stable states. Then the optimized structure was used to calculate the energy with the self-consistent field (SCF) tolerance less than 2×10^{-6} eV·atom^{-1}. ΔH_f ($La_xY_yNi_z$) at 0 K is expressed by the following equation:

$$\Delta H_f(La_xY_yNi_z) = \frac{E(La_xY_yNi_z) - xE(La) - yE(Y) - zE(Ni)}{x + y + z} \quad (4)$$

where $E(La_xY_yNi_z)$, E(La), E(Y) and E(Ni) denote the total energies of $La_xY_yNi_z$ compound, La, Y and Ni at 0 K, respectively. The reference states for pure La, Y and Ni at 0 K are dhcp, hcp and fcc, respectively.

The formation enthalpies of La–Ni and Y–Ni intermetallic compounds were calculated and provided by the Materials Project Database [50, 51] and Zhou et al. [34] as listed in Table 5. The formation enthalpies calculated by Zhou et al. are 5.98–9.74 kJ·mol^{-1} lower than that of the Materials Project Database, and the formation enthalpies calculated in this work are 5.69–11.37 kJ·mol^{-1} higher than that of the Materials Project Database. The differences in calculation results may be caused by different calculation conditions. However, the formation enthalpies calculated in this work are more complete than those from the Materials Project Database and Zhou et al. [34]. In addition, the difference of 5.69–11.37 kJ·mol^{-1} is reasonable compared with the calculated value from Materials Project Database and Zhou et al. [34]. In order to keep the self-consistence of the formation enthalpies, the data in this work are selected as the starting values for optimizing the phase stability parameters. Then the experimental phase relations are used to evaluate the interaction parameters of phases.

Due to the lack of information about the thermodynamic database of the La–Y system, its thermodynamic description was constructed based on the experimental data of Spedding et al. [35] and Wang et al. [36]. The calculated La–Y phase diagram is given in Fig. 11.

The thermodynamic parameters of the constituent binary systems La–Ni [29], Y–Ni [33], and La–Y were employed to extrapolate the thermodynamic description of La–Y–Ni system. This work optimized the interaction parameters of $La_{0.5}Y_{0.5}Ni_5$ and 3R-LaY_2Ni_9 according to the phase equilibria experimental data. The solid solubilities of those intermetallic compounds are also considered, but the optimization mainly depends on phase equilibria. All the

Table 5 Calculated formation enthalpies of intermetallic compounds

Phases	Prototype	Space group	ΔH_f / (kJ·mol^{-1})		
			This work	Materials project [50, 51]	Ref. [34]
La_7Ni_{16}	La_7Ni_{16}	$I\bar{4}2m$	− 21.92	− 33.29	–
3R-La_2Ni_7	Gd_2Co_7	$R\bar{3}m$	− 22.10	–	–
2H-La_2Ni_7	Ce_2Ni_7	$P6_3/mmc$	− 21.24	− 30.39	–
2H-La_5Ni_{19}	Sm_5Co_{19}	$P6_3/mmc$	− 22.88	–	–
$LaNi_5$	$CaCu_5$	$P6/mmm$	− 21.13	− 27.21	–
YNi_2	$TmNi_2$	$Fd\bar{3}m$	− 31.39	− 40.33	− 46.31
3R-YNi_3	Gd_2Co_7	$R\bar{3}m$	− 30.82	− 38.11	− 46.31
2H-Y_2Ni_7	Ce_2Ni_7	$P6_3/mmc$	− 29.91	− 35.60	–
3R-Y_2Ni_7	Gd_2Co_7	$R\bar{3}m$	− 29.57	− 36.76	− 45.35
YNi_5	$CaCu_5$	$P6/mmm$	− 25.59	− 30.78	− 40.52
3R-LaY_2Ni_9	$PuNi_3$	$R\bar{3}m$	− 28.00	–	–

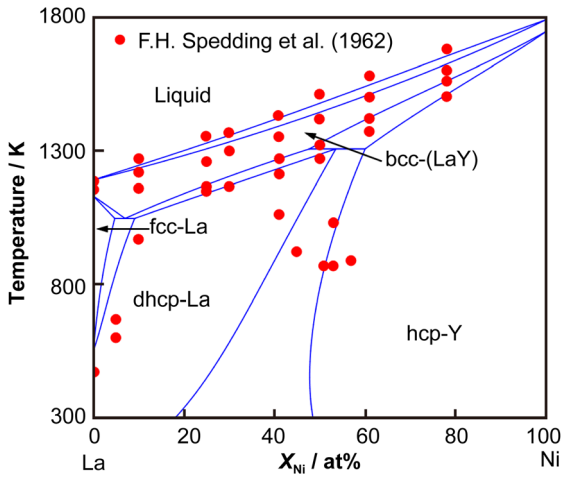

Fig. 11 Calculated La-Y phase diagram with experimental data [35, 36]

experimental data in this work are weighted equally. After several iterations of optimization, the calculated isothermal sections of Ni-rich corner in the La–Y–Ni system at 1273 and 1148 K are shown in Fig. 12. Table 6 shows the thermodynamic models for phases of the La–Y–Ni systems. The calculated phase equilibria were consistent with the experimental data.

4 Conclusion

The phase equilibria of the La–Y–Ni ternary system in Ni-rich corner at 1148 and 1273 K were determined using equilibrated alloys. The results can be summarized as follows:

(1) The 3R-YNi_3, 3R-Y_2Ni_7, 2H-La_5Ni_{19}, 3R-LaY_2Ni_9 and $La_{0.5}Y_{0.5}Ni_5$ phases are found to exist in the La–Y–Ni ternary system at 1273 and 1148 K. When the temperature is below 1273 K, the 2H-La_2Ni_7 phase starts to form. The solubility of the third element in those superlattice compounds increases with the increasing of heat treatment temperature.

(2) The substitution of La for Y would lead to a corresponding increase in the lattice constant and the unit cell volume of 3R-YNi_3. The 3R-LaY_2Ni_9 phase with a larger cell unit volume than 3R-YNi_3 would provide more available interstices for H in the lattice and show higher hydrogen capacity.

(3) Both YNi_5 and $La_{0.5}Y_{0.5}Ni_5$ are $CaCu_5$-type structure. But their lattice parameters are different due to the La solid solubility. There is a miscibility gap between YNi_5 and $La_{0.5}Y_{0.5}Ni_5$.

(4) According to the experimental results, a reasonable thermodynamic description of the La–Y–Ni ternary system

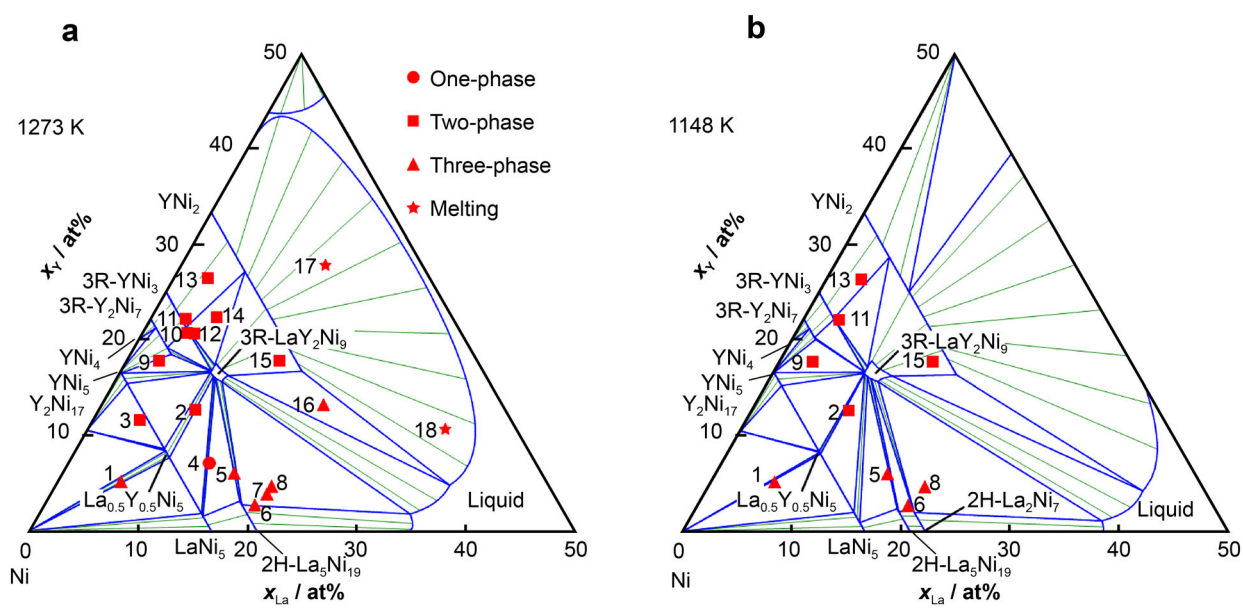

Fig. 12 Isothermal section of Ni-rich corner in La–Y–Ni ternary system at **a** 1273 K and **b** 1148 K

Table 6 Summary of evaluated thermodynamic parameters of La-Y-Ni system

Phases	Model	Parameters	Refs.
Liquid	(La, Y, Ni)	${}^{0}L^{\mathrm{Liquid}}_{\mathrm{La,Ni}} = -114278.224 + 7.1592T$	[29]
		${}^{1}L^{\mathrm{Liquid}}_{\mathrm{La,Ni}} = -63511.8961 + 40.6196T$	[29]
		${}^{0}L^{\mathrm{Liquid}}_{\mathrm{Y,Ni}} = -155496 + 38.932T$	[33]
		${}^{1}L^{\mathrm{Liquid}}_{\mathrm{Y,Ni}} = -52904 + 6.785T$	[33]
		${}^{0}L^{\mathrm{Liquid}}_{\mathrm{La,Y}} = -350$	This work
bcc	(La, Y, Ni, Va)	${}^{0}L^{\mathrm{bcc}}_{\mathrm{La,\ Ni:Va}} = +100000$	[29]
		${}^{0}L^{\mathrm{bcc}}_{\mathrm{Y,Ni:Va}} = -2000$	[33]
		${}^{0}L^{\mathrm{bcc}}_{\mathrm{La,Y:Va}} = +500$	This work
		${}^{1}L^{\mathrm{bcc}}_{\mathrm{La,Y:Va}} = -800$	This work
fcc	(La, Y, Ni, Va)	${}^{0}L^{\mathrm{fcc}}_{\mathrm{La,Ni:Va}} = +100000$	[29]
		${}^{0}L^{\mathrm{bcc}}_{\mathrm{Y,Ni:Va}} = -2000$	[33]
hcp	(La, Y, Ni)	${}^{0}L^{\mathrm{hcp}}_{\mathrm{Y,Ni}} = -2000$	[33]
		${}^{0}L^{\mathrm{hcp}}_{\mathrm{La,Y}} = -9000 + 3.7T$	This work
dhcp	(La, Y, Ni)	${}^{0}L^{\mathrm{dhcp}}_{\mathrm{La,Ni}} = +100000$	[29]
		${}^{0}L^{\mathrm{dhcp}}_{\mathrm{La,Y}} = -450$	This work
Y_2Ni_{17}	Y_2Ni_{17}	${}^{0}G^{\mathrm{Y_2Ni_{17}}}_{\mathrm{Y:Ni}} = 2{}^{0}G^{\mathrm{hcp}}_{\mathrm{Y}} + 17{}^{0}G^{\mathrm{fcc}}_{\mathrm{Ni}} - 386087 + 61.742T$	[33]
YNi_5	$(La, Y)_1Ni_5$	${}^{0}G^{\mathrm{YNi_5}}_{\mathrm{Y:Ni}} = {}^{0}G^{\mathrm{hcp}}_{\mathrm{Y}} + 5{}^{0}G^{\mathrm{fcc}}_{\mathrm{Ni}} - 186706 + 31.187T$	[33]
		${}^{0}G^{\mathrm{YNi_5}}_{\mathrm{La:Ni}} = {}^{0}G^{\mathrm{dhcp}}_{\mathrm{La}} + 5{}^{0}G^{\mathrm{fcc}}_{\mathrm{Ni}} - 80000$	This work
		${}^{0}G^{\mathrm{YNi_5}}_{\mathrm{Y,La:Ni}} = -52600$	This work
YNi_4	Y_1Ni_4	${}^{0}G^{\mathrm{YNi_4}}_{\mathrm{Y:Ni}} = {}^{0}G^{\mathrm{hcp}}_{\mathrm{Y}} + 4{}^{0}G^{\mathrm{fcc}}_{\mathrm{Ni}} - 160555 + 22.979T$	[33]
Y_2Ni_7	$(La, Y)_2Ni_7$	${}^{0}G^{\mathrm{Y_2Ni_7}}_{\mathrm{Y:Ni}} = 2{}^{0}G^{\mathrm{hcp}}_{\mathrm{Y}} + 7{}^{0}G^{\mathrm{fcc}}_{\mathrm{Ni}} - 294966 + 38.77T$	[33]
		${}^{0}G^{\mathrm{Y_2Ni_7}}_{\mathrm{La:Ni}} = 2{}^{0}G^{\mathrm{dhcp}}_{\mathrm{La}} + 7{}^{0}G^{\mathrm{fcc}}_{\mathrm{Ni}} - 160000$	This work
		${}^{0}G^{\mathrm{Y_2Ni_7}}_{\mathrm{Y,La:Ni}} = -176300$	This work
YNi_3	$(La, Y)_1(Y, Ni)_3$	${}^{0}G^{\mathrm{YNi_3}}_{\mathrm{Y:Ni}} = {}^{0}G^{\mathrm{hcp}}_{\mathrm{Y}} + 3{}^{0}G^{\mathrm{fcc}}_{\mathrm{Ni}} - 134358 + 16.477T$	[33]
		${}^{0}G^{\mathrm{YNi_3}}_{\mathrm{La:Ni}} = {}^{0}G^{\mathrm{dhcp}}_{\mathrm{La}} + 3{}^{0}G^{\mathrm{fcc}}_{\mathrm{Ni}} - 60000$	This work
		${}^{0}G^{\mathrm{YNi_3}}_{\mathrm{Y:Y}} = 4{}^{0}G^{\mathrm{hcp}}_{\mathrm{Y}} + 100000$	This work
		${}^{0}G^{\mathrm{YNi_3}}_{\mathrm{Y:Y,Ni}} = -60000$	This work
		${}^{0}G^{\mathrm{YNi_3}}_{\mathrm{La,Y:Ni}} = -107200$	This work
		${}^{1}G^{\mathrm{YNi_3}}_{\mathrm{La,Y:Ni}} = +6000$	This work
		${}^{0}G^{\mathrm{YNi_3}}_{\mathrm{La,Y:Y,Ni}} = -120000$	This work
YNi_2	$(La, Y)_1Ni_2$	${}^{0}G^{\mathrm{YNi_2}}_{\mathrm{Y:Ni}} = {}^{0}G^{\mathrm{hcp}}_{\mathrm{Y}} + 2{}^{0}G^{\mathrm{fcc}}_{\mathrm{Ni}} - 101736 + 8.988T$	[33]
		${}^{0}G^{\mathrm{YNi_2}}_{\mathrm{La:Ni}} = {}^{0}G^{\mathrm{dhcp}}_{\mathrm{La}} + 2{}^{0}G^{\mathrm{fcc}}_{\mathrm{Ni}} - 60000$	This work
		${}^{0}G^{\mathrm{YNi_2}}_{\mathrm{La,Y:Ni}} = -90000$	This work
YNi	Y_1Ni_1	${}^{0}G^{\mathrm{YNi}}_{\mathrm{Y:Ni}} = {}^{0}G^{\mathrm{hcp}}_{\mathrm{Y}} + {}^{0}G^{\mathrm{fcc}}_{\mathrm{Ni}} - 67908 + 6.11T$	[33]
Y_3Ni_2	Y_3Ni_2	${}^{0}G^{\mathrm{Y_3Ni_2}}_{\mathrm{Y:Ni}} = 3{}^{0}G^{\mathrm{hcp}}_{\mathrm{Y}} + 2{}^{0}G^{\mathrm{fcc}}_{\mathrm{Ni}} - 168600 + 37.482T$	[33]
Y_3Ni	Y_3Ni_1	${}^{0}G^{\mathrm{Y_3Ni}}_{\mathrm{Y:Ni}} = 3{}^{0}G^{\mathrm{hcp}}_{\mathrm{Y}} + {}^{0}G^{\mathrm{fcc}}_{\mathrm{Ni}} - 84446 + 14.05T$	[33]
$LaNi_5$	$(La, Y)_1Ni_5$	${}^{0}G^{\mathrm{LaNi_5}}_{\mathrm{La:Ni}} = {}^{0}G^{\mathrm{dhcp}}_{\mathrm{La}} + 5{}^{0}G^{\mathrm{fcc}}_{\mathrm{Ni}} - 168451.2938 + 29.256T$	[33]
		${}^{0}G^{\mathrm{LaNi_5}}_{\mathrm{Y:Ni}} = {}^{0}G^{\mathrm{hcp}}_{\mathrm{Y}} + 5{}^{0}G^{\mathrm{fcc}}_{\mathrm{Ni}} - 86200$	This work
		${}^{0}G^{\mathrm{LaNi_5}}_{\mathrm{La,Y:Ni}} = -72600$	This work

Table 6 continued

Phases	Model	Parameters	Refs.
La_5Ni_{19}	$(La, Y)_5Ni_{19}$	${}^0G^{La_5Ni_{19}}_{La:Ni} = 5{}^0G^{dhcp}_{La} + 19{}^0G^{fcc}_{Ni} - 699079 + 105.053T$	[29]
		${}^0G^{La_5Ni_{19}}_{Y:Ni} = 5{}^0G^{hcp}_{Y} + 19{}^0G^{fcc}_{Ni} - 380000$	This work
		${}^0G^{La_5Ni_{19}}_{La,Y:Ni} = -410000$	This work
La_2Ni_7	$(La, Y)_2Ni_7$	${}^0G^{La_2Ni_7}_{La:Ni} = 2{}^0G^{dhcp}_{La} + 7{}^0G^{fcc}_{Ni} - 264471 + 37.486T$	[29]
		${}^0G^{La_2Ni_7}_{Y:Ni} = 2{}^0G^{hcp}_{Y} + 7{}^0G^{fcc}_{Ni} - 160000$	This work
		${}^0G^{La_2Ni_7}_{La,Y:Ni} = -140000$	This work
$LaNi_3$	La_1Ni_3	${}^0G^{LaNi_3}_{La:Ni} = {}^0G^{dhcp}_{La} + 3{}^0G^{fcc}_{Ni} - 119954.235 + 16.657T$	[29]
La_7Ni_{16}	La_7Ni_{16}	${}^0G^{La_7Ni_{16}}_{La:Ni} = 7{}^0G^{dhcp}_{La} + 16{}^0G^{fcc}_{Ni} - 689590 + 79.955T$	[29]
La_2Ni_3	La_2Ni_3	${}^0G^{La_2Ni_3}_{La:Ni} = 2{}^0G^{dhcp}_{La} + 3{}^0G^{fcc}_{Ni} - 148509 + 11.297T$	[29]
LaNi	La_1Ni_1	${}^0G^{LaNi}_{La:Ni} = {}^0G^{dhcp}_{La} + {}^0G^{fcc}_{Ni} - 56450.823 + 0.265T$	[29]
La_7Ni_3	La_7Ni_3	${}^0G^{La_7Ni_3}_{La:Ni} = 7{}^0G^{dhcp}_{La} + 3{}^0G^{fcc}_{Ni} - 197310.48 + 1.054T$	[29]
La_3Ni	La_3Ni_1	${}^0G^{La_3Ni}_{La:Ni} = 3{}^0G^{dhcp}_{La} + {}^0G^{fcc}_{Ni} - 67410.7 - 0.702T$	[29]
$La_{0.5}Y_{0.5}Ni_5$	$(La, Y)_{0.5}(La, Y)_{0.5}Ni_5$	${}^0G^{La_{0.5}Y_{0.5}Ni_5}_{La:Y:Ni} = {}^0G^{dhcp}_{La} + {}^0G^{hcp}_{Y} + 10{}^0G^{fcc}_{Ni} - 300000$	This work
		${}^0G^{La_{0.5}Y_{0.5}Ni_5}_{La:La:Ni} = 2{}^0G^{dhcp}_{La} + 10{}^0G^{fcc}_{Ni} - 240000$	This work
		${}^0G^{La_{0.5}Y_{0.5}Ni_5}_{Y:La:Ni} = {}^0G^{hcp}_{Y} + {}^0G^{dhcp}_{La} + 10{}^0G^{fcc}_{Ni} + 30000$	This work
		${}^0G^{La_{0.5}Y_{0.5}Ni_5}_{Y:Y:Ni} = 2{}^0G^{hcp}_{Y} + 10{}^0G^{fcc}_{Ni} - 260000$	This work
LaY_2Ni_9	$(La, Y, Ni)_1(La, Y, Ni)_2(La, Y, Ni)_9$	${}^0G^{LaY_2Ni_9}_{La:Y:Ni} = {}^0G^{dhcp}_{La} + 2{}^0G^{hcp}_{Y} + 9{}^0G^{fcc}_{Ni} - 500000$	This work
		${}^0G^{LaY_2Ni_9}_{La:La:Ni} = 3{}^0G^{dhcp}_{La} + 9{}^0G^{fcc}_{Ni} - 226800$	This work
		${}^0G^{LaY_2Ni_9}_{Y:Y:Ni} = 3{}^0G^{hcp}_{Y} + 9{}^0G^{fcc}_{Ni} - 241600$	This work
		${}^0G^{LaY_2Ni_9}_{La:Y:Y} = {}^0G^{dhcp}_{La} + 11{}^0G^{hcp}_{Y} - 8000$	This work
		${}^0G^{LaY_2Ni_9}_{Y:Y:Y} = 12{}^0G^{hcp}_{Y} + 8104.8$	This work
		${}^0G^{LaY_2Ni_9}_{La:Y:La} = 10{}^0G^{dhcp}_{La} + 2{}^0G^{hcp}_{Y} - 1000$	This work
		${}^0G^{LaY_2Ni_9}_{La:La:La} = 12{}^0G^{dhcp}_{La} + 9000$	This work
		${}^0G^{LaY_2Ni_9}_{La:Ni:Ni} = {}^0G^{dhcp}_{La} + 11{}^0G^{fcc}_{Ni} - 81540$	This work
		${}^0G^{LaY_2Ni_9}_{Ni:Y:Ni} = 2{}^0G^{hcp}_{Y} + 10{}^0G^{fcc}_{Ni} - 227500 + 10T$	This work
		${}^0G^{LaY_2Ni_9}_{Ni:Ni:Ni} = 12{}^0G^{fcc}_{Ni} + 10000$	This work
		${}^0G^{LaY_2Ni_9}_{La,Ni:Y:Ni} = -11850$	This work
		${}^0G^{LaY_2Ni_9}_{La:Y,Ni:Ni} = -18000$	This work
		${}^0G^{LaY_2Ni_9}_{La:La,Y:Ni} = -22000$	This work
		${}^0G^{LaY_2Ni_9}_{La:Y:La,Ni} = -182000$	This work
		${}^1G^{LaY_2Ni_9}_{La:Y:La,Ni} = -26870$	This work
		${}^0G^{LaY_2Ni_9}_{La:Y:Y,Ni} = -199300$	This work
		${}^0G^{LaY_2Ni_9}_{La:La,Y:La,Ni} = -826050$	This work
		${}^1G^{LaY_2Ni_9}_{La:La,Y:La,Ni} = -27050$	This work

was established. The calculated phase equilibria showed a good agreement with the experimental data.

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Declarations

Conflict of interests The authors declare that they have no conflict of interest.

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